

Tyrone Marhguy

Computer Engineering — Computer Architecture / RTL / FPGA / ASIC / Verification

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SUMMARY

Computer Engineering student building from transistor-level CMOS and physical design through RTL, FPGA systems, CPU/GPU microarchitecture, networking, formal verification, and ASIC implementation. Built custom processors, a programmable 3D GPU, low-latency Ethernet/ITCH datapaths, SRAM/MAC blocks, and upstream EDA fixes.

EDUCATION

University of Pennsylvania

B.S.E. Computer Engineering

Philadelphia, PA

Aug. 2024 – May 2028

- Coursework: Computer Organization & Design, VLSI, Digital Logic, Embedded Systems

EXPERIENCE

Hardware & Firmware Engineer

Vero Electric

May 2026 – Aug. 2026

Philadelphia, PA

- Developed embedded C/C++ fault-detection firmware and a **BQ79616 16-channel battery-monitoring front end** for a **240 kWh / 120 kW** system; closed PCB ERC with **0 errors** and debugged bring-up with lab instruments.

Software Engineer Intern

Aragorn AI

May 2026 – Aug. 2026

Remote

- Shipped production backend services and REST APIs; resolved cross-service failures blocking releases and hardened service boundaries through architecture/code reviews.

SELECTED HARDWARE PROJECTS

Tomato — Custom 32-bit Polymorphic CPU | Website | GitHub

SystemVerilog / FPGA / KiCad / Formal

- Architected a custom 32-bit CPU around two programmable **LUT3 planes + 8-way carry/flag select (524,288 ALU configurations)**, a three-source datapath, and microcoded control; built a working FPGA computer with assembler and TomatoOS over HDMI.
- Verified up to **130B Verilator vectors** plus formal 1/8/32-bit proofs; fabricated the discrete ALU board and continued the complete 74xx/discrete machine toward physical bring-up.

Out-of-Order 64-bit RISC-V CPU (RV64IM) | GitHub

SystemVerilog / RISC-V / AXI4-Lite

- Built RV64IM with A/C bring-up (riscv64xO3 core) and a **2-wide fetch/decode/issue/commit** pipeline, gshare predictor + BTB + RAS, AXI4-Lite memory, and Verilator/cocotb/riscv-tests verification with Yosys/OpenLane2 feasibility checks.

Pineapple GPU P1 — Standalone Programmable 3D GPU | GitHub

Verilog / Artix-7 / DVI

- Implemented vertex/fragment shaders, triangle rasterization, depth, texture sampling, and double-buffered **320x180** output; rendered interactive pineapple and cube scenes on a Nexys A7 using a FOSS Yosys/nextpnr flow.

Low-Latency FPGA Networking & NASDAQ ITCH | UDP/IP | ITCH

SystemVerilog / RMII / cocotb

- Built a deterministic **100 Mbps RMII/MAC/IPv4/UDP** pipeline and an RTL ITCH 5.0 parser + 512-entry BRAM limit book; closed networking at **100 MHz** and verified ITCH against a Python/cocotb golden model over a **10M-message** regression with zero BBO mismatches.

ASIC / Full-Custom Datapaths | MAC | SRAM

Sky130 / 22 nm HP / SPICE

- Designed a BF16 multiply-to-FP32-accumulate MAC with LibreLane/OpenLane hardening and a transistor-level **16x4 6T SRAM** with StrongARM sensing; automated NGSpice sweeps and achieved simulated functional readback to **4.571 GHz**.

OPEN-SOURCE EDA

OpenROAD, Verilator, LibreLane, OpenFPGA | Merged upstream

C++ / Python / EDA Infrastructure

- Merged upstream fixes to production EDA tools: repaired an OpenROAD LEF58/ODB parser, corrected Verilator Linux peak-memory accounting, and restored LibreLane/Yosys compatibility/reporting; LibreLane fixes shipped in releases.

TECHNICAL SKILLS

Architecture / RTL: SystemVerilog/Verilog, VHDL, CPU/GPU/SoC microarchitecture, pipelining, OoO concepts, ISA/datapath design, memory systems, CDC

Verification / ASIC: UVM, SVA, cocotb, formal, Verilator; Vivado, Yosys, OpenROAD, LibreLane/OpenLane, Sky130, synthesis, STA, P&R, DRC/LVS, SPICE, KiCad, Electric VLSI

Programming / Interfaces: Python, C/C++, Embedded C, Tcl, Bash, Linux; Ethernet/RMII/IPv4/UDP, AXI4-Lite, UART, SPI, I2C